MHF4U Advanced Functions - Unit 1 Practice Test (Version 1)

Functions and Rates of Change

Name: _____ Date: _____ Time: 90 minutes | Total: 60 marks

This is an original practice test based on the supplied Unit 1 index. It is not an official Ontario school board test. A scientific or graphing calculator and graph paper may be used.

Instructions: Answer every question. Show your reasoning for full marks. Use correct interval notation, label any sketches, and include units and a concluding sentence for context questions. Formal differentiation rules are not required.

Part A - Knowledge and Understanding (15 marks)

1. [5 marks] Let `f(x) = sqrt(6 - x)/(x + 2)`.

a. State the domain in interval notation. [3]

b. Explain both restrictions used to obtain your answer. [2]

2. [5 marks] For `g(x) = -2|x - 3| + 4`, state:

a. the vertex and direction of opening; [1]

b. the x-intercepts; [2]

c. the domain and range in interval notation. [2]

3. [5 marks] For `h(x) = 1/(x + 3) - 2`, state:

a. the vertical and horizontal asymptotes; [2]

b. the domain and range in interval notation; [2]

c. the y-intercept. [1]

Part B - Thinking (15 marks)

4. [5 marks] The point `(9, 3)` lies on `y = f(x)`, where `f(x) = sqrt(x)`. Define `g(x) = -2f(3(x + 1)) + 4`.

a. Find the corresponding point on g. Show the input mapping. [2]

b. State the domain and range of g. [2]

c. Describe the transformations from `f` to g. [1]

5. [5 marks] Let `f(x) = (x - 1)^2 + 2`, with the restricted domain `x >= 1`.

a. Find `f^(-1)(x)`. [2]

b. State the domain and range of the inverse. [2]

c. Verify your inverse by calculating `f(f^(-1)(6))`. [1]

6. [5 marks] Consider the piecewise function

```
           x + 1       if x < 0
p(x) =     1 - x^2     if 0 <= x <= 2
           3           if x > 2
```

a. Sketch the graph, showing open and closed endpoints. [2]

b. State the range in interval notation or set notation. [1]

c. Is the function continuous at `x = 0`? At `x = 2`? Explain. [2]

Part C - Communication (15 marks)

7. [5 marks] A student says: "For `q(x) = f(2(x - 3)) + 1`, the graph of `f` shifts 6 units right because `2 x 3 = 6`." Explain the student's error. Give the correct horizontal scale and shift, then map the point `(4, -1)` on `f` to a point on `q`.

8. [5 marks] A sensor records the following position values (metres):

Time t (s)	**1.9**	**2.0**	**2.1**
Position s(t) (m)	9.79	10.00	10.19

a. Calculate the average rate of change over `[1.9, 2.0]` and `[2.0, 2.1]`. [2]

b. Estimate the instantaneous rate at `t = 2.0` using a centred interval. [1]

c. Explain, with units, how the two average rates support your estimate. [2]

9. [5 marks] A student claims that `f^(-1)(x) = 1/f(x)` for every function. Use `f(x) = 2x - 5` to correct the claim. Find both expressions, explain their different meanings, and verify the inverse using one composition.

Part D - Application (15 marks)

10. [5 marks] A parking garage charges $6 for any stay of up to 2 hours. After 2 hours, it adds $3 per additional hour, prorated for part of an hour. The maximum stay is 8 hours. Let `C(t)` be the charge for a stay of `t` hours, where `0 < t <= 8`.

a. Write a piecewise rule for `C(t)`. [2]

b. Find `C(4.2)`. [1]

c. Find the average rate of change in cost from `t = 1` to `t = 5`, and interpret it. [2]

11. [5 marks] During its first 4 seconds, a ball's height is modelled by `H(t) = -5t^2 + 20t + 1` metres.

a. Find the average rate of change in height from `t = 1` to `t = 3`. [2]

b. Estimate the instantaneous rate at `t = 1` using the centred interval `[0.99, 1.01]`. [2]

c. Explain why the answers to a and b can differ. [1]

12. [5 marks] A school club models daily revenue by `R(p) = -2(p - 10)^2 + 200` dollars, where p is the ticket price in dollars and `0 <= p <= 20`.

a. Calculate the average rate of change of revenue from `p = 9` to `p = 11`. [2]

b. Estimate the instantaneous rate at `p = 10` using the centred interval `[9.9, 10.1]`. [1]

c. State the price giving maximum modelled revenue, the revenue, and why the rate results fit that maximum. [2]